

'Simone' Memory Game Activity

Hello research participant!

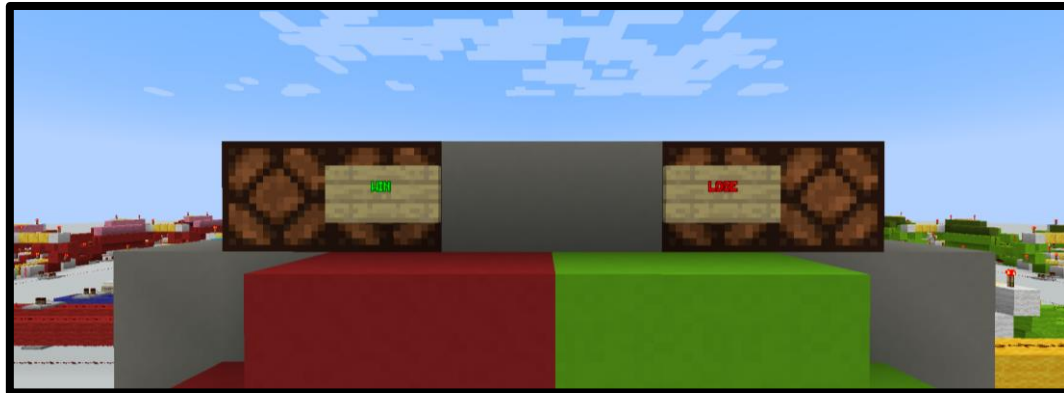
This activity involves fixing a **minigame** in Minecraft.



Researching the Self-perception of Computational Thinking skills learnt through Game-based learning

...and whether incorporating principles of Gameful Design can enhance it.

'Simone' Memory Game Activity



Requirements:

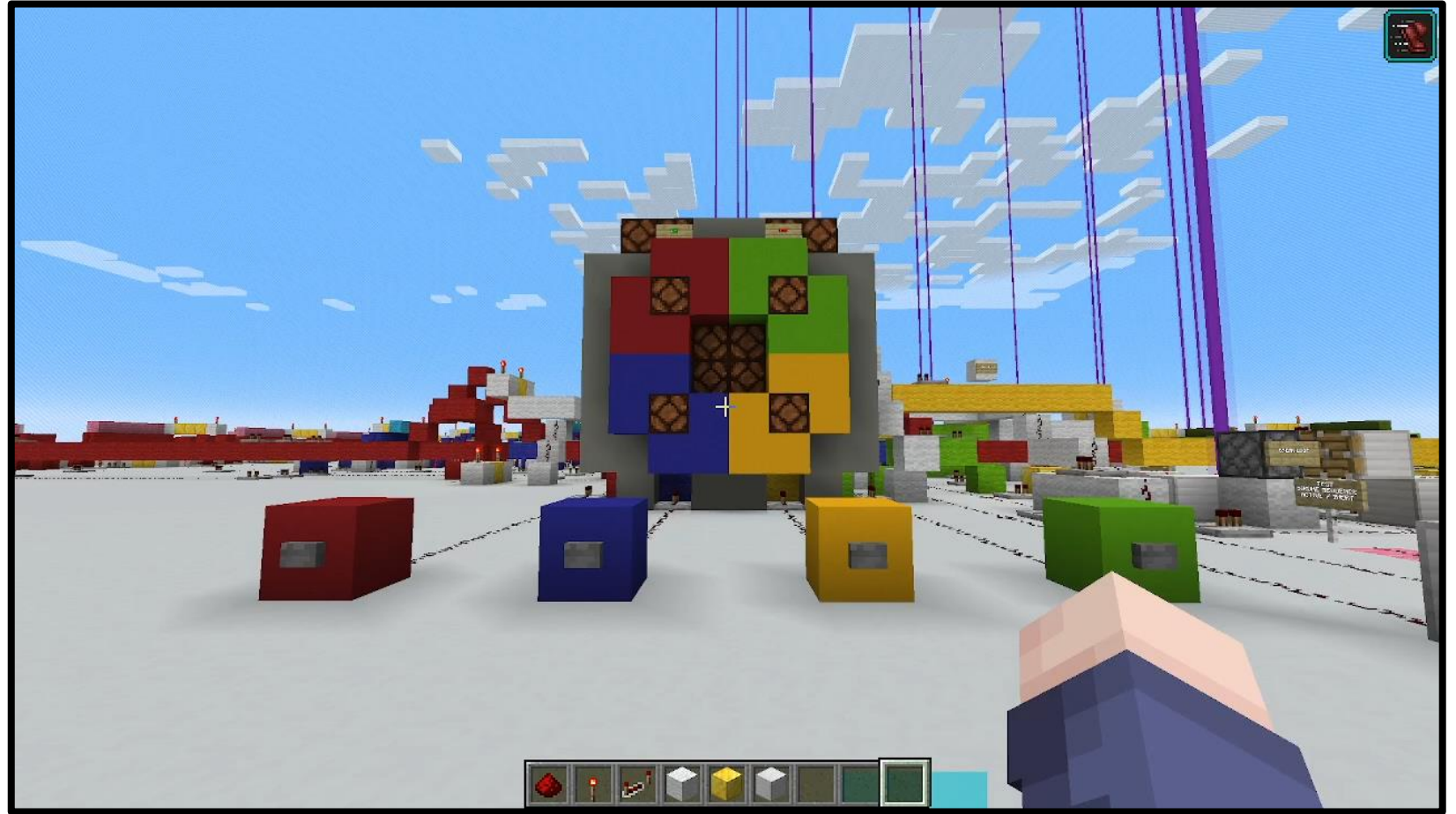
- **'Simone'** creates a 5-colour sequence, (very slowly) then outputs a flash.
- **Player** can input a 5-colour sequence (a little slowly)
- If sequence is **correct**, display **win**.
- If sequence is **incorrect**, display **lose**.

'Simone' Memory Game Activity

Here's the minigame working!

It is **slow**.

Remember to wait for the central **flash** before inputting.



(If viewing through pdf, video doesn't work)

'Simone' Memory Game Activity

Some bits are broken! Areas needing fixing are highlighted in **Purple**.

Nearby signs with **purple text** describe the purpose of the required component.

If you can **cover** the **purple wool** without destroying it, you're on the right track to the needed component.



Testing Components



Components can be **Tested!**

Pressing the **corresponding button** on **Pink** wool will confirm whether the component is correct or not.

If testing incidentally occurs without a button press, ignore it.

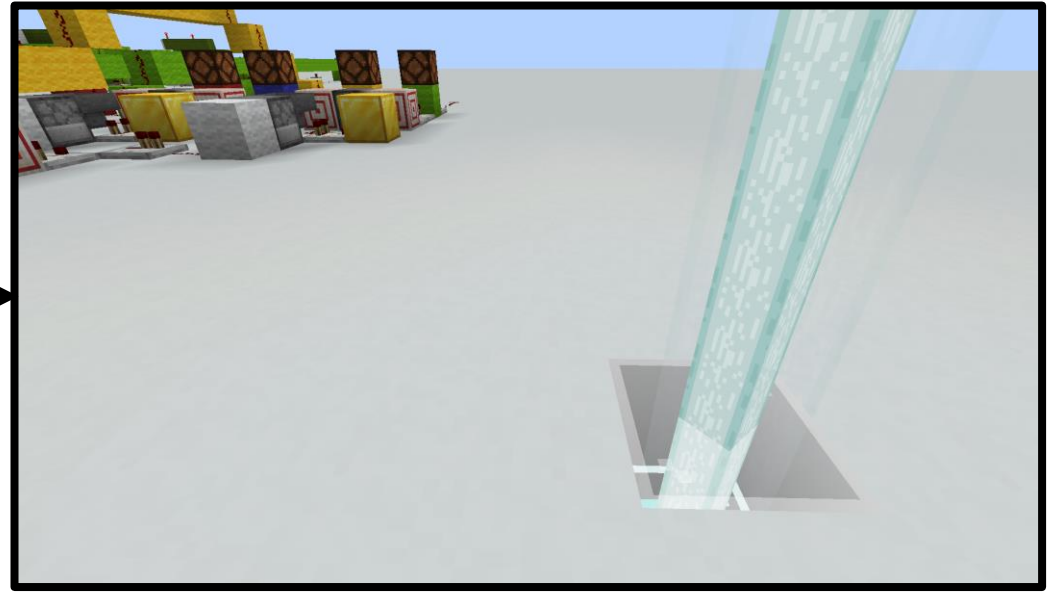


Remember to **disable testing** before playing the minigame.

Testing Components

Once you've implemented a missing component, feel free to break the nearby signs and replace the purple glass with... any glass colour you like!

Try to only do this if the component is tested and correct!

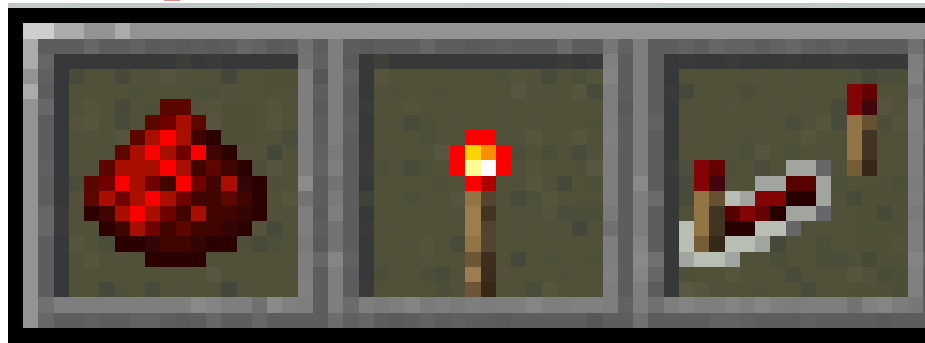


Minecraft Redstone

REDSTONE DUST – **Transports** a redstone signal.

REDSTONE TORCH – **Outputs** a redstone signal **unless** a signal is applied to it through a block.

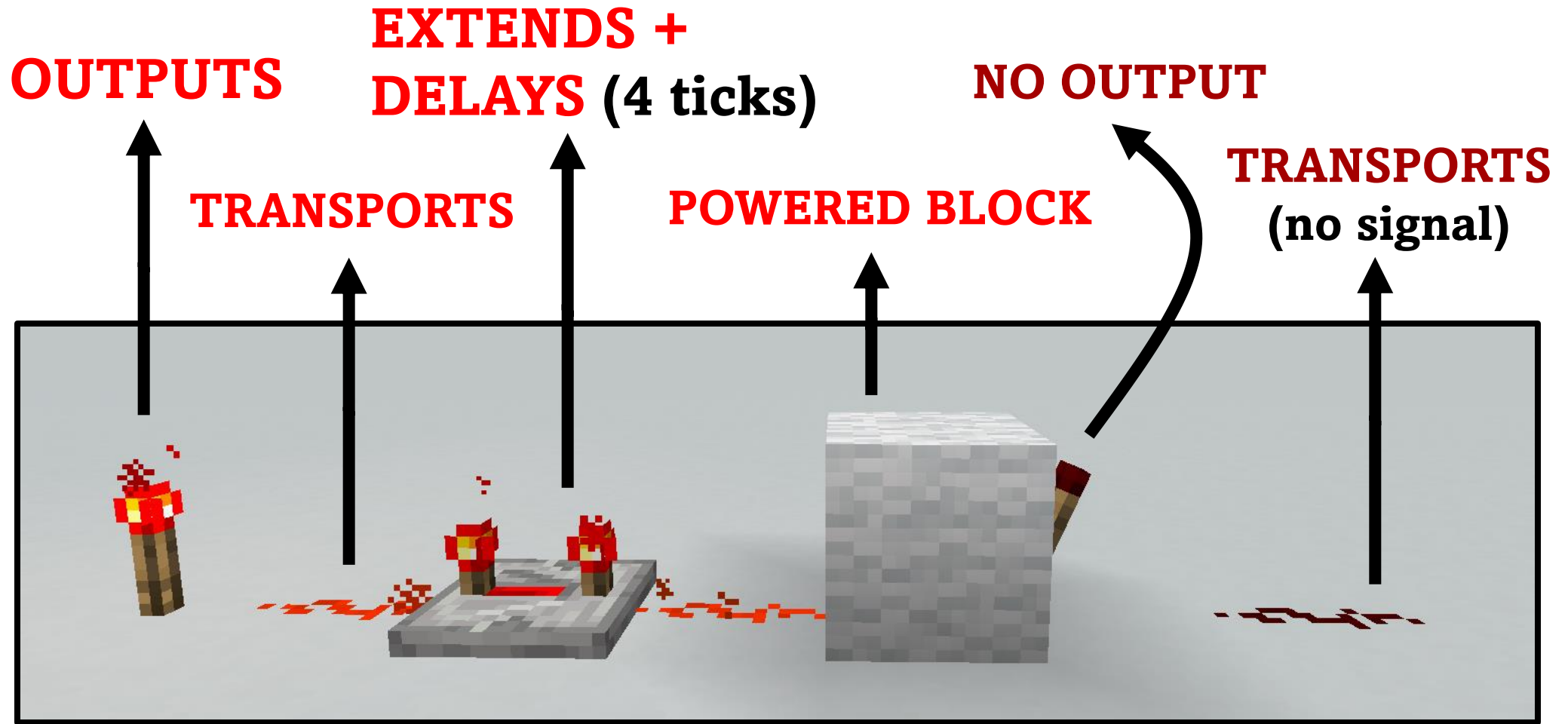
REDSTONE REPEATER – **Delays** and **Extends** a redstone signal in **one direction**.



**BUILDING
BLOCKS**



Minecraft Redstone



Tool Blocks

After successfully testing a type of component for the first time, you may **unlock** a **Tool Block**.

These are **not necessary**, but may help with implementing additional components.



Feel free to experiment with them!

Questionnaire!

Whenever you're finished with the activity, whether you've completed it or not, please answer the following **questionnaire**:

<https://forms.office.com/e/ymBkxvLWhU>

You can also access it via
this button in-game



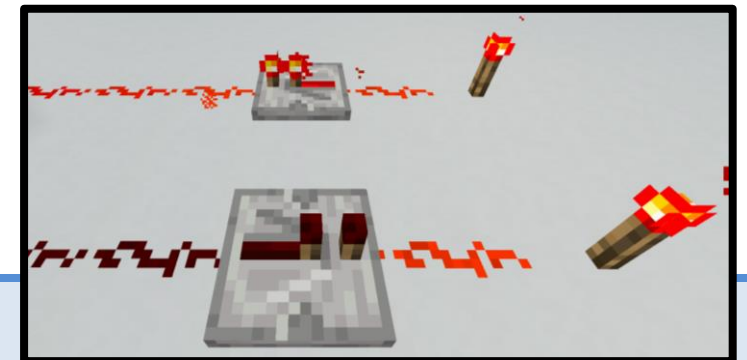
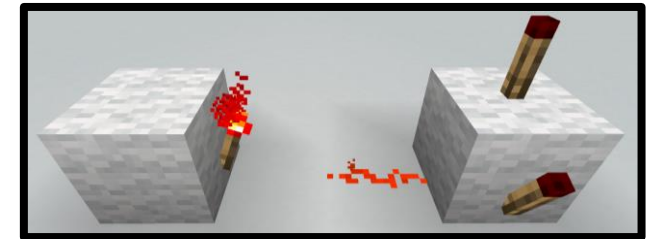
Good luck!



- Aim to **cover** the **purple wool**!
- **Test** your components!



- **REDSTONE TORCH** – **Outputs** a redstone signal **unless** a signal is applied to it through a block.
- **REDSTONE REPEATER** – **Delays** and **Extends** a redstone signal in **one direction**.



Feel free to ask any questions at any point :)